Day 2: TCP/IP Suite Theory (1 hour): What is the TCP/IP Suite? Layers of TCP/IP: Application, Transport, Internet, and Network Access

Tasks

1.Open Wireshark, capture network traffic while you refresh the Flipkart homepage, and identify one packet each belonging to the Application, Transport, Internet, and Network Access layers by examining their headers.

ANS: STEP 1: Open **Wireshark** on your computer.

STEP 2: Select your active Wi-Fi or Ethernet network interface.

STEP 3: Click **Start Capture**.

STEP 4: Open Google Chrome and visit https://www.flipkart.com.

STEO 5: Refresh the Flipkart homepage.

STEP 6: Return to Wireshark and stop the capture.

STEP 7: Apply the filter tcp or http to examine packets.

STEP 8: Select individual packets and expand their headers to identify the layers.

Packet Identification Table

TCP/IP Layer	Protocol / Header	What to Identify
Application	HTTP / TLS	HTTP request or encrypted HTTPS data
Transport	TCP	Source port, destination port
Internet	IP	Source IP and destination IP
Network Access	Ethernet / Wi-Fi	Source MAC and destination MAC

Example Observation

Layer	Example Packet Details
Application	TLS – Client Hello
Transport	TCP – Destination Port 443
Internet	IPv4 – Source and Destination IP
Network Access	Ethernet II – Source and Destination MAC

2.Create a table comparing the OSI and TCP/IP models, listing all layers side-by-side, and for each layer, give a real-world example from any app you use (e.g., WhatsApp message, Zomato order).

ANS:

OSI Model (7 Layers)	TCP/IP Model (4 Layers)	Real-Life Example (WhatsApp)
7. Application	Application	I type and send "Hello" on WhatsApp.
6. Presentation	Application	My message is converted into a suitable format and encrypted.
5. Session	Application	WhatsApp keeps the connection between two users.
4. Transport	Transport	My message is divided into small parts and sent.
3. Network	Internet	IP address helps find the receiver's device.
2. Data Link	Network Access	Wi-Fi sends data from my phone to the router.
1. Physical	Network Access	Wi-Fi signals carry data through the air.

Main Differences

OSI Model	TCP/IP Model
It has 7 layers.	It has 4 layers.
It explains how networks work.	It explains how internet communication works.
It is a reference model.	It is used for real internet communication.
Each layer has a separate job.	Some layers are combined.

Real-Life Example: Sending a WhatsApp Message

Suppose I send **"Hello"** to my friend.

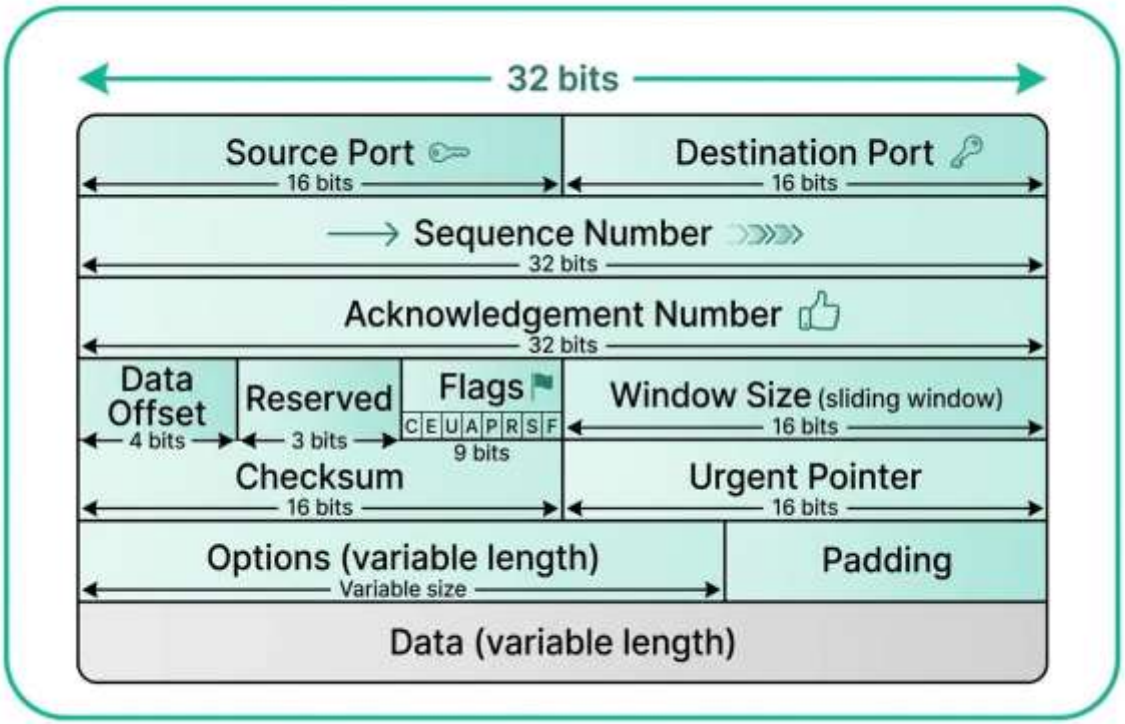
1. **Application:** I open WhatsApp and type Hello.
2. **Presentation:** The message is encrypted for privacy.
3. **Session:** WhatsApp manages the communication between us.
4. **Transport:** The message data is prepared and sent.
5. **Network:** IP addresses help deliver the data to my friend's phone.

6. **Data Link:** My Wi-Fi sends the data to the router.
7. **Physical:** Wi-Fi signals carry the data through the air.

**3.Given a screenshot of a TCP segment header (provided by your trainer), label the fields that belong to the Transport layer and explain what each field does in 1-2 lines.

Hint: Look for fields like Source Port, Destination Port, Sequence Number, etc.**

ANS:



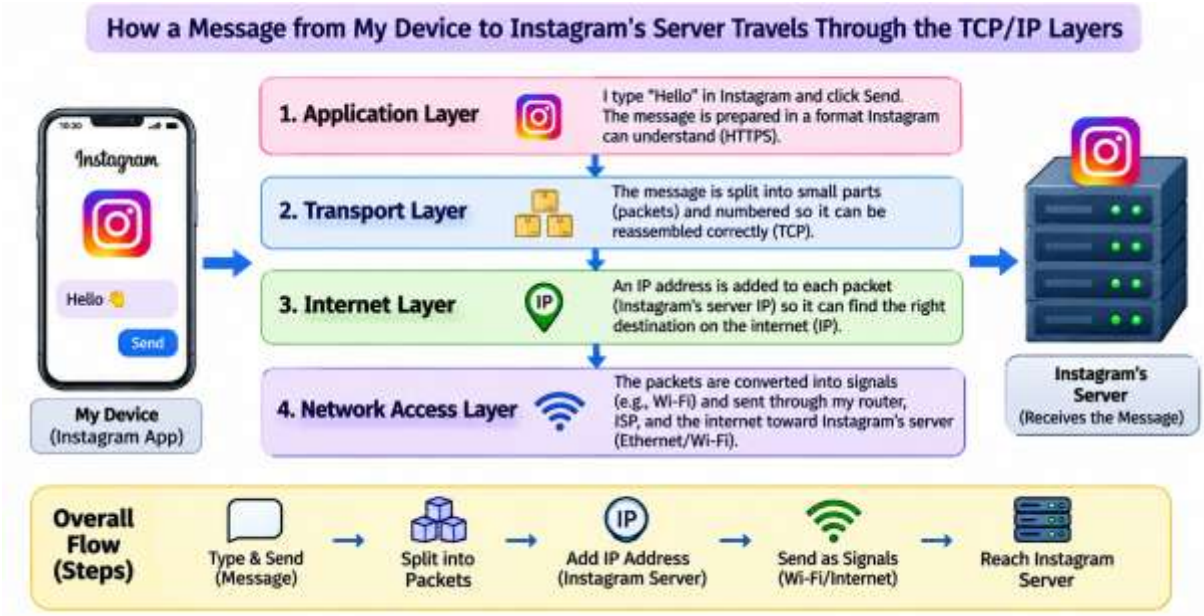
TCP Header Field	What Does It Do? (Easy Words)
Source Port	Shows which application is sending the data.
Destination Port	Shows which application should receive the data.
Sequence Number	Keeps data in the correct order.
Acknowledgment Number	Tells the sender which data has been received.
Header Length	Shows the size of the TCP header.
Flags	Control the connection, such as starting or ending communication.
Window Size	Tells how much data the receiver can accept at a time.
Checksum	Checks whether the data has errors or is damaged.
Urgent Pointer	Identifies urgent data that needs special attention.
Options	Contains extra information for TCP communication.

Real-Life Example:
When I open WhatsApp and send a message to my friend:

1. Source Port identifies the sending application connection.
2. Destination Port identifies the receiving service.
3. Sequence Number keeps data in order.
4. Acknowledgment Number confirms received data.
5. Checksum checks for errors during transmission.

4.Draw a diagram showing how a message sent from your device to Instagram's server passes through each TCP/IP layer, and annotate what happens at each step using simple app-related terms (e.g., 'message split into packets', 'IP address added', etc.).

ANS:



1. **Application Layer:** I type "Hello" and click Send on Instagram.
2. **Transport Layer:** The message is divided into small packets.
3. **Internet Layer:** The destination IP address is added.
4. **Network Access Layer:** Data travels through Wi-Fi or mobile internet to Instagram's server.

5.Pick any one packet you captured in Wireshark and write which TCP/IP layer it belongs to, how you identified it, and what real-world action in an app (like sending a WhatsApp message or booking a ticket on BookMyShow) it represents.

ANS:

Packet Details

Item	Answer
Packet	TCP Packet
TCP/IP Layer	Transport Layer
Protocol	TCP
Source Port	53218 (Example)
Destination Port	443 (Example)

How I Identified It

1. I opened Wireshark and captured network traffic.
2. I selected one TCP packet.
3. I checked the packet details.
4. I found Source Port and Destination Port.
5. These fields helped me identify the Transport Layer.

Real-Life Example

When I send a message on WhatsApp, TCP helps transfer data between devices and keeps the data in the correct order.